



First Name :
Last Name :

Test N°1

Exercise 1:

Which of the following are not valid identifiers (Check the box for non-valid IDs)?

- | | | | |
|-------------------------|-------------------------------------|-------------------------|-------------------------------------|
| • Hauteur1 | ☐ | • lArgEur_2 | ☐ |
| • épaisseur | ☒ | • POIDS-TOTAL | ☒ |
| • Niveau_d'eau | ☒ | • Longueur_ | ☐ |
| • longueur_plus_2 | ☐ | • 2_fois_longueur | ☒ |
| • Adresse.Email | ☒ | • #include | ☒ |

Exercise 2:

What is the result of each expression?

- | | |
|----------------------------|--------------------------|
| • $1 + 4 * 5 - 2 = 19$ | • $(1 + 4) * 5 - 2 = 23$ |
| • $1 + 4 * (5 - 2) = 13$ | • $1 + (4 * 5) - 2 = 19$ |
| • $(1 + 4) * (5 - 2) = 15$ | • $(1 + 4 * 5) - 2 = 19$ |

Exercise 3:

Consider the following algorithm declarations:

$a \leftarrow 8$ (Integer), $b \leftarrow 3$ (Integer), $c \leftarrow 2$ (LongInt), $y \leftarrow 4.5$ (Real)

Give the type and value of each of the following expressions:

Expression	Value	Type	Expression	Value	Type
• $a + c$	10	LongInt	• $a + y$	12.5	Real
• $a \text{ MOD } b + c$	4	LongInt	• $a < b$	false	Boolean
• $a \geq b$	true	Boolean	• NOT ($a > c$)	false	Boolean
• $(a > b) \text{ AND } (c < y)$	true	Boolean	• $(a < 10) \text{ OR } (b > 5)$	true	Boolean
• $c + 2 * (a > b)$	Error	X	• $(a - 8) \text{ AND } (c - 2)$	Error	X

Exercise 4:

Write an algorithm that calculates the final grade, knowing that the final grade is calculated using the following formula:

$$\text{Final}_{\text{Grade}} = \text{CC}_{\text{Grade}} \times 40\% + \text{EFS}_{\text{Grade}} \times 60\%, \quad \text{where: } \text{CC}_{\text{Grade}} = \frac{\text{DW}_{\text{Grade}} + \text{PW}_{\text{Grade}}}{2}$$

```

Algorithm Calc_Final_Grade;
Var
    Final, CC, EFS, DW, PW: Real;
Begin
    Write("Give the values of DW, PW, EFS");
    Read(DW, PW, EFS);
    CC ← (DW+PW)/2;
    Final ← (CC*0.4)+(EFS*0.6);
    Write(" Final Grade is: ", Final);
End.

```